3.9 Angle Relationship Proofs – Part 2

<table>
<tr><th colspan="6">Lesson Introduction</th></tr>
<tr><td>Benchmarks</td><td colspan="2">MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.</td><td>Learning Target</td><td colspan="2">• I can prove relationships using parallel lines and their transversals.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.</td><td colspan="3">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="5">• How can you prove relationships using parallel lines and their transversals?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">Students will complete a Card Sort activity of eight proofs of varying difficulty to practice proving relationships using parallel lines and their transversals.</td></tr>
<tr><td rowspan="2">Component Summary</td><td>3.9.1 Warm-Up (~5 min.)</td><td>Notice and Wonder about ant hills (SE p. 170)</td><td>3.9.2 Activity (35-40 min.)</td><td colspan="2">Complete two-column proofs involving parallel lines and transversals by sorting given statements and reasons (SE p. 170)</td></tr>
<tr><td>3.9.3 Wrap-Up (~5 min.)</td><td colspan="4">Students summarize their learning (SE p. 174)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
• Parallel Lines
• Transversal
• Congruent
• Supplementary</td><td colspan="2">• (Optional) Chart Paper
• (Optional) Markers
• (Optional) Laminated Proofs and dry erase markers</td><td colspan="2">• The teacher will need to have printed the Statement and Reason Cards, and have them separated into sets for each group. It is recommended to have group sizes of 2-3 students so the sets should be assembled depending on your class size.
• If implementing the Take Turns routine, divide the cards up in each set for students to be assigned to each role.</td></tr>
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<tr>
<td rowspan="4">Teacher Tips</td>
<td>Common Misconceptions</td>
<td>Things to Consider</td>
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Students may mix up which cards are statements and which are reasons.
</td>
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This lesson centers on a Card Sort activity. Consider using one of the following methods to complete this activity.

Traditional Card Sort
Gallery Walk
Chart Completion
Laminated Proofs

If students struggle completing proofs, consider having a sheet available to hand out where some of the steps have been completed for them.

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<tr>
<td>Literacy and Language Routines</td>
<td>Mathematical Thinking and Reasoning (MTR) Standard</td>
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<td>Take Turns
For the Card Sort in 3.9.2 Activity, set students up to use the Take Turns routine. Decide ahead of time how best to divide the sorting cards between the partners to ensure that both students are participating and one student doesn’t automatically take over the activity.</td>
<td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students who participate in effortful learning, both individually and with others, are able to analyze the problem in a way that makes sense, given the task. They will be able to ask questions that will help them solve the task. This will build perseverance by allowing them to modify methods, as needed while solving a challenging task.</td>
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<td>Differentiate Instruction</td>
<td colspan="2">The Card Sort activity offers many opportunities for differentiation. The whole class can work through the proofs in the same way, or different groups can be assigned to work through the activity in different ways, such as small group instruction. Students can also be assigned only some of the proofs instead of completing all of them. Use data from the previous lesson to determine appropriate differentiation to meet the needs of your students.</td>
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<td colspan="3">Lesson Notes:</td>
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3.9.1 Warm-Up: Notice and Wonder

Component Overview: *Students are provided with a photo taken of ant hills on a brick pathway and are asked what they notice and wonder.*

3.9.1 Warm-Up: Notice and Wonder

A photo of ant hills taken in Gainesville, Florida is shown.

1. What do you notice?
   Answers will vary. The ant hills lie on the creases formed by the straight lines between the bricks.

2. What do you wonder?
   Answers will vary. Do that ant hills have tunnels underneath that connect them and do they run parallel to the creases in the bricks.

3. Write a caption for the photo.
   Answers will vary. Ants Are Great Geometry Students

Do you know someone who is a bricklayer or landscape architect? Consider inviting a guest speaker to join your class virtually or in person. Students can ask the guest speaker questions.
Consider polling the class to share students' creative captions for the photo.

3.9.2 Activity: Card Sort

Component Overview: *3.9.2 Activity positions students to strengthen their knowledge of the relationships between angles and their transversal(s) and skills by completing proofs. They will demonstrate their knowledge of the different relationships as well as their understanding of what can be deduced from given and known information.*

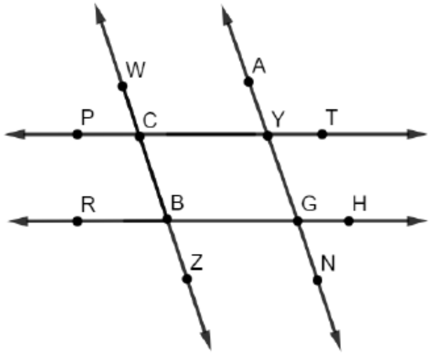
3.9.2 Activity: Card Sort

1. With your group, sort through the statements and reasons for each proof. Then, write each proof in the tables provided.

Proof A

Given: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$, $\angle WCY \cong \angle YGH$

Prove: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$



Statement	Reason
$\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	*Given*
$\angle WCY \cong \angle YGH$	*Given*
$\angle WCY \cong \angle AYT$	*If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.*
$\angle AYT \cong \angle YGH$	*Transitive Property of Congruence*
$\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	*If two lines are intersected by a transversal so that the corresponding angles are congruent, then the lines are parallel.*

There are several ways to consider implementing this activity.

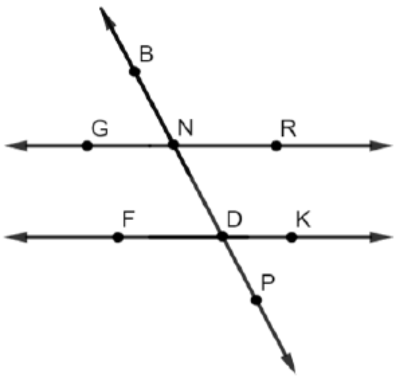
Option 1: Traditional Card Sort – suggested group size of 3.
Option 2: The teacher can assign students certain cards, and then have the students do a Gallery Walk. The students can be instructed to do the Card Sort, then write an incomplete proof on chart paper. The students can then circulate in groups to write in the next statement or reason. Then, the group that sorted the cards can check the proof.
Option 3: The empty proofs can be put on chart paper, and students circulate filling in one statement or reason until all the proofs are complete.
Option 4: The teacher copies the blank proofs and laminates them. The students use dry erase markers to complete the proof, and trade with another group to have the proof evaluated.

Consider developing a checklist for your use as you observe students working on the proofs.

Proof B

Given: $\angle GND$ and $\angle PDK$ are supplementary

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$



Statement	Reason
$\angle GND$ and $\angle PDK$ are supplementary	Given
$m\angle GND + m\angle PDK = 180°$	Definition of Supplementary Angles
$\angle GND \cong \angle BNR$	Vertical Angles are Congruent
$m\angle GND = m\angle BNR$	Definition of Congruence
$m\angle BNR + m\angle PDK = 180°$	Substitution Property of Equality
$\angle BNR$ and $\angle PDK$ are supplementary	Definition of Supplementary Angles
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	If two lines are intersected by a transversal so that the consecutive exterior angles are supplementary, then the lines are parallel.

Notice and Wonder
For each proof, have students think about the following:

- *What do you notice?*
- *What is the same?*
- *What is different?*
- *What does that make you wonder or think about this proof and how it differs from the first one you completed?*
- *Will this always be true? Why or why not?*

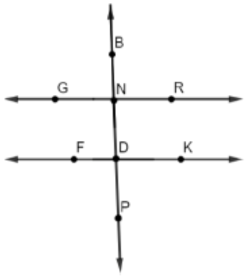
If students struggle with completing these proofs, consider having a sheet available to hand out where some of the steps have been completed for them.

3.9.2 Activity: Card Sort (continued)

Proof C

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$, $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

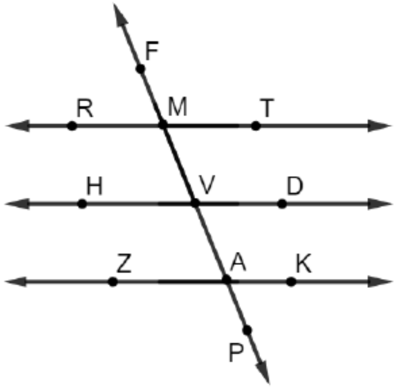
Prove: $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$



Statement	Reason
$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	*Given*
$\angle GNB$ *is a right angle*	*Definition of Perpendicular Lines*
$m\angle GNB = 90°$	*Definition of Right Angle*
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*Given*
$\angle GNB \cong \angle FDN$	*If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.*
$m\angle GNB = m\angle FDN$	*Definition of Congruence*
$m\angle FDN = 90°$	*Substitution Property of Equality*
$\angle FDN$ *is a right angle*	*Definition of Right Angle*
$\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	*Definition of Perpendicular Lines*

When sharing the correct answers for 3.9.2 Activity, consider having groups present their findings, requiring that they also include their explanation and rationale for their answer choices.

3.9.2 Activity: Card Sort (continued)

Proof D	
Given: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$, $\angle TMV \cong \angle AVD$ Prove: $\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$	

Statement	Reason
$\overleftrightarrow{RT} \parallel \overleftrightarrow{ZK}$	*Given*
$\angle TMV \cong \angle KAP$	*If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.*
$\angle TMV \cong \angle AVD$	*Given*
$\angle KAP \cong \angle AVD$	*Transitive Property of Congruence*
$\overleftrightarrow{HD} \parallel \overleftrightarrow{ZK}$	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.*

Consider using some of the time during 3.9.2 Activity to pull small groups of students to provide additional support based on their performance with this component.

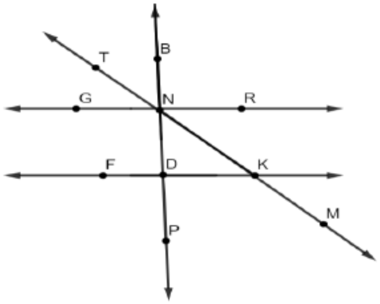
3.9.2 Activity: Card Sort (continued)

For an additional challenge, consider having students write the proof(s) in a different format, such as a flowchart proof or a paragraph proof.

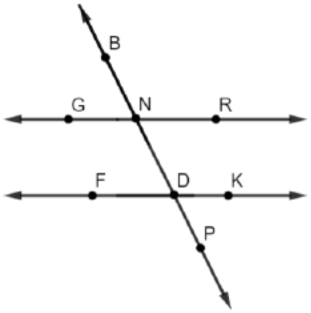
Proof E

Given: $\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$ and $\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$

Prove: $\angle DKM$ and $\angle GNT$ are supplementary



Statement	Reason
$\overleftrightarrow{BP} \perp \overleftrightarrow{GR}$	*Given*
$\angle BNR$ *is a right angle*	*Definition of Perpendicular Lines*
$m\angle BNR = 90°$	*Definition of a Right Angle*
$\overleftrightarrow{BP} \perp \overleftrightarrow{FK}$	*Given*
$\angle NDK$ *is a right angle*	*Definition of Perpendicular Lines*
$m\angle NDK = 90°$	*Definition of Right Angle*
$m\angle BNR = m\angle NDK$	*Substitution Property of Equality*
$\angle BNR \cong \angle NDK$	*Definition of Congruence*
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.*
$\angle DKM$ *and* $\angle GNT$ *are supplementary*	*If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.*

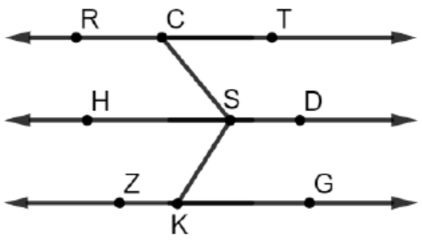
Proof F	
Given: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$ Prove: $\angle GND$ and $\angle PDK$ are supplementary	

Statement	Reason
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*Given*
$\angle GND$ and $\angle FDN$ are supplementary	*If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.*
$m\angle GND + m\angle FDN = 180°$	*Definition of Supplementary*
$\angle FDN \cong \angle PDK$	*Vertical Angles are Congruent*
$m\angle FDN = m\angle PDK$	*Definition of Congruence*
$m\angle GND + m\angle PDK = 180°$	*Substitution Property of Equality*
$\angle GND$ and $\angle PDK$ are supplementary	*Definition of Supplementary Angles*

3.9.2 Activity: Card Sort (continued)

Proof G

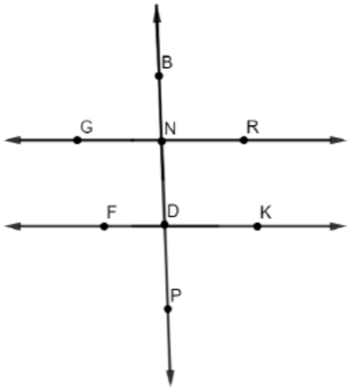
Given:
$\angle TCS \cong \angle HSC$,
$\angle HSK \cong \angle SKG$
Prove: $\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$



Statement	Reason
$\angle TCS \cong \angle HSC$	*Given*
$\overleftrightarrow{RT} \parallel \overleftrightarrow{HD}$	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*
$\angle HSK \cong \angle SKG$	*Given*
$\overleftrightarrow{HD} \parallel \overleftrightarrow{ZG}$	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*
$\overleftrightarrow{RT} \parallel \overleftrightarrow{ZG}$	*Transitive Property of Parallel Lines*

Students may not be clear about the transitive property of parallel lines. The transitive property of parallel lines states that if $\overleftrightarrow{RT}$ is parallel to $\overleftrightarrow{HD}$, and $\overleftrightarrow{HD}$ is parallel to $\overleftrightarrow{ZG}$, then $\overleftrightarrow{RT}$ is parallel to $\overleftrightarrow{ZG}$.
Pause the class or small groups to clarify, as needed.

3.9.2 Activity: Card Sort (continued)

Proof H	
Given: $\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ and $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$ Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	
Statement	**Reason**
$\overleftrightarrow{GR} \perp \overleftrightarrow{BP}$ $\overleftrightarrow{FK} \perp \overleftrightarrow{BP}$	*Given*
$\angle BNR$ *is a right angle.* $\angle NDK$ *is a right angle.*	*Definition of Perpendicular Lines*
$m\angle BNR = 90°$ $m\angle NDK = 90°$	*Definition of a right angle*
$m\angle BNR = m\angle NDK$	*Transitive Property of Equality*
$\angle BNR \cong \angle NDK$	*Definition of Congruence*
$\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$	*If two lines are intersected by a transversal so that corresponding angles are congruent, then the two lines are parallel.*

How is Proof H similar to Proof C? How are they different?

3.9.3 Wrap-Up: Cool Down

Component Overview*: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

3.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about what you've learned about angle relationship proofs. Include things that are easy to remember and common errors to watch out for.
   Answers will vary.